17 Mobilization and Collection of HSCs in Children

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17.1 Introduction

Collecting or harvesting hematopoietic stem cells (HSCs) from children is a challenge, not only because children have physiological and anatomical differences but also because the psychological, legal, and ethical concerns in minors differ from those in adult donors. In addition, parents and/or legal guardians have to be addressed in all issues and can only assent and not consent to the procedure. The main difference to the adult setting is the small body weight (bw), the difficulties in venous access, especially in the leukapheresis setting, and the need for red blood cell (RBC) substitution.

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17.2 Patient Selection and Pre-Procedure Preparation

17.2.1 Patient Selection and Testing

In children, the indications for autologous HSC harvesting is well-established (Passweg et al. 2014). Using children in the allogeneic setting as donors is a completely different issue (Bitan et al. 2016). Children should not donate HSCs if a comparable adult volunteer HSC donor is available, if the indication for the stem cell therapy is not first line, or if the therapy is experimental (Sheldon 2004; Zinner 2004).

Patient pre-procedure analyses follow the same local legal requirements as in adults for blood counts, blood group, clinical chemistry, and infectious disease testing. It may extend in allogeneic sibling donors for parameters that are specific to the transplantation indication, e.g., hemoglobin electrophoresis.

To perform HSC harvesting in children, physicians and nurses must be experienced in the care of children and knowledgeable about the normal age-dependent physiological parameters, like vital signs, growth, and psychological and motor development, and should be trained in the communication with children, parents, and/or their legal guardians (Anthias et al. 2016).

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17.2.2 Bone Marrow (BM) vs. Peripheral Blood Stem Cells (PBSCs) and Risk Analysis

The main graft resources are BM and PBSCs. The basic techniques are quite similar to those used in adults. For BM collection, the iliac crests or, in extremely small children, the tibia is used. For harvesting HSCs from the PB, leukapheresis is used with the same apheresis systems as in adults.

A study from the European Group for Blood and Marrow Transplantation (EBMT) Pediatric Diseases Working Party describes which factors influence the safety of HSC collection. In this prospective evaluation, 453 pediatric donors were included. The children donated either BM or PBSCs according to center policy. A large variability in approach to donor issues was observed between the participating centers. Significant differences were observed between BM and PBSC donors regarding pain, need for RBC support, duration of hospitalization, and iron supplement; however, differences between the groups undergoing BM vs. PBSC donation preclude direct risk comparisons between the two procedures. The most common adverse event was pain, mainly reported by older children after BM harvesting but also observed after central venous catheter (CVC) placement for PBSC collection. With regard to severe adverse events, one patient developed pneumothorax with hydrothorax after CVC placement for PBSC collection. The risk of allo-transfusion after BM harvesting was associated with a donor age of <4 years and a BM harvesting volume of >20 mL/kg. Children <4 years were at a higher risk than were older children for RBC support after BM harvesting, and there was a higher risk of complications from CVC placement before apheresis. It was concluded that PBSC and BM collection are both safe procedures in children (Styczynski et al. 2012).

17.2.3 Children as Allogeneic Donors

Pediatric-aged donors vary widely in their ability to assent or consent to the risks of a donation procedure. There are key regulations and ethical imperatives, which must be addressed in deciding which donation procedure is appropriate for minors (van Walraven et al. 2013). In order to have general guidance, in 2010, the American Academy of Pediatrics published a recommendation on this issue. The authors strongly recommend the inclusion of the pediatric donor in all decision-making processes to the extent that they are capable. An independent chaperon should stand as the minor's advocate to not only protect the rights of the donor but also help prevent any delay of the donation procedure (Chan and Tipoe 2013).

The decision to consider a minor family donor, especially in inherited diseases, is complicated due to the fact that phenotypically healthy or minor symptomatic siblings with a mild carrier status might be eligible to donate to the severely ill recipient. One simple example is a sibling with thalassemia minor for a recipient with a thalassemia major (Biral et al. 2008). There are many other major diseases, including primary immunodeficiencies, chronic granulomatous disease, or sickle cell disease, where carriers are used as HSC donors. Potential family sibling donors with medical or psychological problems should not be considered as donors and therefore should not be HLA-typed (Bitan et al. 2016).

17.3 Bone Marrow Harvesting

Extending from Chap. 15, the collection of HSCs from the BM is historically the oldest technique. Multiple punctures of the iliac crest are performed under general anesthesia by experienced physicians. The bone marrow is harvested by aspirations through adequately dimensioned needles. In extremely small children, and if the iliac crest is anatomically not suitable for punctures, then the aspirations could also be performed by punctures of the proximal tibia.

For successful HCT, it is necessary to obtain enough progenitor cells during the BM harvesting procedure. Most centers use multiple aspirations of maximum 2 mL BM, whereas others use

few larger amounts of aspirations for BM harvesting (20–100–250 mL). It could be shown that the latter methods result in comparable grafts for transplantation (Witt et al. 2016). For some young donors with anatomical limitations or in diseases where a suitable donor should be used for more than one recipient, a minimally harming procedure is warranted for bone marrow harvesting (Biral et al. 2008).

More recently, adult donors have received granulocyte colony-stimulating factor (G-CSF) because stimulated BM is richer in HSCs and therefore results in quicker engraftment (Ji et al. 2002). However, data for pediatric donors are ambiguous (Frangoul et al. 2007; Furey et al. 2018). A recent study has shown that a dose of 3–5 × 10^6 $CD34^+$ hematopoietic progenitor cells (HPCs)/kilogram of recipient body weight was the optimal $CD34^+$ cell dose infused to attain graft-versus-host disease (GVHD)/relapse-free survival in children with an matched sibling donor (MSD) while constraining donor side effects.

17.4 Peripheral Blood Stem Cell Harvesting

17.4.1 Mobilization and Preparation

For mobilization of HPCs into the PB, the longest experience exists with G-CSF in combination with chemotherapy in the autologous setting, but plerixafor has also been reported in case series and in the MOZAIC (Multicenter International Study of Oxaliplatin/Fluorouracil/Leucovorin in the Adjuvant Treatment of Colon Cancer) study, a two-arm phase I/II study, as being safe for the use in children in combination with standard G-CSF mobilization (Chambon et al. 2013; Morland et al. 2020). The dose proposed in this study is 240 μg/kg bw 8–12 h before apheresis (Karres et al. 2020). In two case series, plerixafor is reported as feasible and safe in haploidentical and allogenic donors with an unfavorable donor/recipient bw ratio (Kurnakova et al. 2021; Zubicaray et al. 2021).

As in adults, leukapheresis should be performed if meaningful numbers of $CD34^+$ HPCs are mobilized in the peripheral blood, to achieve the minimal threshold of 2–5 × 10^6/kg recipient with a minimum number of procedures (Fritsch et al. 2010).

17.4.2 Vascular Access

Vascular access can be frequently achieved with only peripheral venous access lines (Witt et al. 2008). For central access, Hickman catheters are usually sufficient, and temporary Shaldon catheter placement is only required in a minority of patients (Doberschuetz et al. 2019). Alternative line management with arterial access is also possible (Goldstein 2012; Even-Or et al. 2013; Hunt et al. 2013).

17.4.3 Apheresis Techniques

PBSCs are harvested by leukapheresis in extremely small children even below 6 kg body weight and have been described since the 1990s of the last century (Kanold et al. 1994; Klingebiel et al. 1995; Diaz et al. 1996). Special experience and techniques are required to perform safe leukapheresis procedures in pediatric patients using apheresis systems, which are constructed for use in adults. Priming of these systems with saline and citrate as for adults may cause anemia and possibly dilution coagulopathy in children below, e.g., 15–30 kg and 10 kg body weight, respectively, due to the large extracorporeal volume of the apheresis systems (ca. 160–220 mL) (Moog 2010). Blood warmers, if used, take an additional 50 mL (Pasko et al. 2023). The expected blood loss for the procedure should therefore be calculated (Witt et al. 2007). This has to be done individually to decide whether a priming of the set is needed. In most of the newest versions of the apheresis systems, an algorithm guides the user through this pediatric priming procedure. For priming only irradiated and leukodepleted packed RBCs should be used. This can be supplemented by plasma products as applied for intrauterine exchange transfusion, if coagulopathy is anticipated.

After completion of a primed apheresis, tube rinsing with saline is usually omitted to prevent circulatory overload from the priming red cell unit in the patient.

Priming for low-body-weight children includes the adjustment of apheresis machine pump rates, if less than 10 mL/min inlet flow is required. A possible workaround for apheresis machine setting limitations is by priming with an artificial body weight setting, e.g., 50 kg and by reducing this to the patient's body weight immediately before connecting the patient. Caution should be exercised for continuous pump operations, as this may be technically limited to 5 mL/min. In addition, blood flow speed can be too slow to achieve timely anticoagulation.

For anticoagulation, citrate is mostly used in extremely small children with an initial anticoagulant citrate dextrose solution A (ACD-A) rate of 1:12 and increases in case of clumping and low blood flow with a high risk of hypocalcemic side effects (Pasko et al. 2023). Alternatively, ACD-A may be supplemented by heparin, e.g., 5000 IE per 500 mL ACD-A that allows rates of up to 1:22 (Salazar-Riojas et al. 2015).

17.4.4 Apheresis Monitoring

To avoid hypocalcemia-related adverse effects, meticulous, ionized calcium monitoring by blood gas analysis is recommended (Kreuzer et al. 2011; Maitta et al. 2014). Calcium substitution is frequently required and recommended to be applied by a separate line, as low flow rates considerably increase the risk of intravenous coagulation by calcium excess compared to adults.

Volume management is a crucial aspect of pediatric stem cell apheresis to ensure patient safety and optimize the collection of HSCs. Adequate volume management involves careful consideration of blood flow rates, processed volumes, replacement fluids, and monitoring of patient hemodynamics.

Patient compliance to the procedure is frequently sufficiently high to avoid sedation, especially if parents are present. However, if children display discomfort, this is typically tied to central venous pressure variations that impair collection efficacy. In these instances, sedation as required is recommended, e.g., using propofol (Devasia et al. 2021).

Key Points

- Pediatric donors can safely donate HSCs if an experienced team is performing the harvesting procedure.
- Donors below 4 years of age have a higher risk of harvesting-associated complications. With BM harvesting, they have a higher need for RBC support and there is a higher risk of complications from CVC placement before apheresis.
- Minors should only be recruited as HSC donors if no medically equivalent histocompatible adult is available for donation and if there is a reasonable likelihood that the recipient will benefit from the procedure.
- An informed consent (child assent) for HSC donation has to be obtained by the legal guardians and from the pediatric donor. A donor advocate with expertise in pediatric development should be appointed for all minors who are considered as potential HSC donors.
- Long-term follow-up data should be collected to help determine the actual medical and psychological benefits and risks of child donors.

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